
3.2.1 Living organisms vary and this variation is influenced by genetic and environmental factors.

Investigating variation	Variation exists between members of a species. The need for random sampling, and the importance of chance in contributing to differences between samples. The concept of normal distribution about a mean. Understanding mean and standard deviation as measures of variation within a sample. Candidates will not be required to calculate standard deviation in questions on written papers. Candidates should be able to analyse and interpret data relating to interspecific and intraspecific variation.
Causes of variation	Similarities and differences between individuals within a species may be the result of genetic factors, differences in environmental factors, or a combination of both. Candidates should appreciate the tentative nature of any conclusions that can be drawn relating to the causes of variation.
